



Cboe FI FIX Drop Copy Gateway Specification

Version 1.1.2

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1 Introduction

The Cboe FI FIX Drop Copy Gateway provides clients with a feed of executed trades from the Cboe FI platform over FIX Protocol. The feed is intended to be used by clients for reconciliation or for providing trade data to back office and settlement systems.

1.1 Hours of Operation

All times are listed in Eastern Time (ET) which uses EST (UTC-05) during fall/winter and EDT (UTC-04) during spring/summer months when daylight savings time is observed.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Start	7:30 AM	7:30 AM	7:30 AM	7:30 AM	7:30 AM	N/A	N/A
Stop	5:00 PM	5:00 PM	5:00 PM	5:00 PM	5:00 PM	N/A	N/A

For Holidays, Cboe FI follows the SIFMA recommendation for Government Securities.

1.2 Network Connectivity, Security, and Encryption

Cboe FI supports a number of connectivity options including the Internet, leased lines, VPN, and third party networks. Third party networks include Radianz, GuavaTech, Yipes, and TNS. The client should be prepared to provide Cboe FI source IP addresses for all servers that will be establishing connectivity to Cboe FI. Cboe FI's connectivity team will help our clients establish connectivity through a standard on-boarding process.

Clients should consult the latest copy of the Cboe FI [connectivity manual](#) on the Cboe FI website for additional information.

For Production, Cboe FI requires using secure socket layer (SSL) for any client that chooses to connect to Cboe FI Drop Copy Gateway through unrestricted Internet. Our server in our Certification environment does not require use of SSL.

1.3 Symbology

Cboe FI uses the below table for trading instrument symbology. All products available for trading on Cboe FI are On-the Run ("OTR") US Treasuries.

Symbol	Description	Type
US2YT	2-Year Note	TNOTE
US3YT	3-Year Note	TNOTE
US5YT	5-Year Note	TNOTE
US7YT	7-Year Note	TNOTE
US10YT	10-Year Note	TNOTE
US20YT	20-Year Bond	TBOND
US30YT	30-Year Bond	TBOND

2 General FIX Message Structure

The Standard Header and Standard Trailer are required on all FIX messages. MsgType (Tag 35) is part of the header.

2.1 Supported FIX Version

Cboe FI supports the FIX Protocol Version 4.2. Readers of this document should be familiar with common FIX Protocol standards, and refer to standard FIX Protocol as documented at <https://www.fixtrading.org/>.

For developer convenience, Custom FIX Tags are highlighted in blue in the Cboe FI document. Cboe FI has the following custom FIX Tags:

Tag	Field Name
6997	AccountID
6998	SessionID

2.2 Standard Header

Tag	FieldName	Req'd	Description
8	BeginString	Y	Should always be "FIX.4.2". Cboe FI Drop Copy uses FIX 4.2.
9	BodyLength	Y	
35	MsgType	Y	See Supported Message Type section for accepted message types.
34	MsgSeqNum	Y	Message sequence numbers reset to '1' at the start of each trading day.
49	SenderCompID	Y	
56	TargetCompID	Y	
43	PossDupFlag	N	Used to indicate a possible retransmission of the sequence number.
97	PossResend	N	Used to indicate a possible retransmission of the message under a new sequence number.
52	SendingTime	Y	GMT Date/Time message was sent. • The value should ideally contain milliseconds. • It must be within 2 minutes of the Cboe FI system time to not be rejected. • The Cboe FI FIX Gateway uses PTP time synchronization. It is strongly recommended the sender use either PTP or NTP time synchronization.

2.3 Standard Trailer

Tag	FieldName	Req'd	Description
10	Checksum	Y	Integer byte count of message length without the CheckSum field. Follows FIX protocol standard.

2.4 Message Types

The following message types are supported by the Cboe FI FIX Gateway.

Message Type	Message	Originator
0	Heartbeat	Cboe FI and Client
1	Test Request	Cboe FI and Client
2	Resend Request	Cboe FI and Client
3	Reject	Cboe FI
4	Sequence Reset	Cboe FI and Client
5	Logout	Cboe FI and Client
8	Execution Report	Cboe FI
A	Logon	Cboe FI and Client

2.5 FIX Tag-Specific Notes

- **Tag 58 (Text):** Freeform text that is for informational purposes only. Clients should not be coding to the exact value of this field. Cboe FI retains the right to change the values sent in this field.

3 Administrative Messages

3.1 Logon Request (From Client)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="A"
553	Username	N	Cboe FI username.
554	Password	N	Cboe FI password.
98	EncryptMethod	Y	Always 0.
108	HeartBtInt	Y	Client heartbeat interval, expressed in seconds.
141	ResetSeqNumFlag	N	Indicates the intention to reset sequence numbers on both sides of the connection. Setting this field will prevent resequencing from occurring. Client Logon messages with 141=Y are HIGHLY discouraged other than on initial daily logon. The reason is that it's possible for messages to be sent to but never received by a client during a network outage. If, in this scenario, a client resequences correctly on logon, they will be resent the missed messages. However, if the client's logon message contains 141=Y, resequencing will not occur, and the missed messages will not be sent.

3.2 Logon Response (From Cboe FI)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="A"
98	EncryptMethod	Y	Always 0.
108	HeartBtInt	Y	Client heartbeat interval, expressed in seconds.
141	ResetSeqNumFlag	N	Echoed value from the Logon message sent by clients. Not set if a client does not set the field in their Logon message.

3.3 Heartbeat (From Cboe FI and Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="0"
112	TestReqID	N	Only included if Heartbeat is in response to a Test Request.

3.4 Test Request (From Cboe FI and Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="1"
112	TestReqID	Y	

3.5 Resend Request (From Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="2"
7	BeginSeqNo	Y	
16	EndSeqNo	Y	0 indicates +infinity.

In response to a Resend Request, Cboe FI will only respond to clients with trades. On resent trades, Cboe FI does not use the PossDupFlag Tag (Tag 43).

3.6 Sequence Reset/Gap Fill (From Cboe FI and Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="4"
36	NewSeqNo	Y	Next expected sequence number.
123	GapFillFlag	N	Valid values are 'Y' or 'N'. The use of 'Y' is required for common resequencing. 'N' is not recommended, and should only be used in emergency situations that require manual intervention. 'N' causes the SeqNo of the SequenceReset message to be ignored, creating a high possibility of message loss.

3.7 Logout (From Cboe FI and Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="5"
58	Text	N	Freeform text to indicate the reason for the logout.

4 Cboe FI Execution Report Messages

****Note:** On occasion the system may send one or more duplicate trade messages (all fields precisely the same). All clients must be able to handle/process a duplicate trade message.

4.1 Execution Report (From Cboe FI to Clients)

Tag	FieldName	Req'd	Description
35	MsgType	Y	MsgType="8"
1	Account	N	Alphanumeric text field meant to contain 3rd-party account ID. This will only be populated if sent in the Account Tag (Tag 1) on a New Order Single message.
11	ClOrdID	Y	Client-generated Order ID. For takers the field contains the ClOrdID (Tag 11) from the New Order Single message (35=D). For makers, the field contains the QuoteID field (Tag 117) concatenated with the QuoteEntryID field (Tag 299) of the Quote message (35=S). For trades not directly initiated by the client (manual adjustment trades, for example) the field may be set to 0.
17	ExecID	Y	Cboe FI Execution ID. Execution IDs for Cboe FI are indefinitely unique and never reused. The Execution ID field is only composed of alphanumeric characters and underscores.
20	ExecTransType	Y	Always 0.
37	OrderID	Y	Exchange Generated OrderID.
39	OrdStatus	Y	Always 2 (Filled).
40	OrdType	Y	Always 2.
38	OrderQty	Y	
44	Price	Y	
59	TimeInForce	Y	Value will always be 0 regardless of the TimeInForce on the original order.
15	Currency	Y	Always 'USD'.
54	Side	Y	1 = Buy 2 = Sell
64	SettlementDate	Y	Settlement date of the trade in YYYYMMDD format.
75	TradeDate	Y	Trade date in YYYYMMDD format.
55	Symbol	Y	The symbol for the Treasury instrument. Refer to the symbology section of the spec for example Cboe FI symbology.

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22	SecurityIDSource	Y	Always 1.
48	SecurityID	Y	CUSIP for the instrument.
76	ExecBroker	Y	Y = Indicates aggressive trade (Liquidity Consumers Only) N = Indicates passive trade (Liquidity Providers Only) CBOEUST = Otherwise (manually entered trade)
151	LeavesQty	Y	OrderQty (Tag 38) - CumQty (Tag 14). Notwithstanding the above, any remaining quantity of a Market Maker quote will be automatically removed from the orderbook.
14	CumQty	Y	
150	ExecType	Y	Always F (Trade).
6	AvgPx	Y	
31	LastPx	Y	Executed price.
32	LastShares	Y	Executed quantity.
58	Text	N	Freeform string text.
60	TransactTime	Y	
6997	AccountID	Y	Account responsible for the trade.
6998	SessionID	Y	Session responsible for the trade.

4.2 Optional: Execution Report Acknowledgement (From Clients to Cboe FI)

Clients can choose to have acknowledgements required for execution report messages sent by the Cboe FI FIX Drop Copy Server. The functionality is optional, and users looking to have their sessions configured for the feature should contact the Cboe FI Trade Desk.

- 1 Clients enabled for the functionality are expected to respond back to an Execution Report from Cboe FI with a duplicate Execution Report containing the same details within 5 seconds of receipt. In the event the client does not respond in the 5 second window, or sends an incomplete Execution Report message, Cboe FI will send an email alert to the client notifying them of the missing execution report acknowledgement.

Support

Please, e-mail questions or comments regarding this specification to:
fitradedesk@cboe.com or tradedesk@cboe.com.

Revision History

Version	Date	Description
1.0.0	03/10/2021	Initial version.
1.1.0	11/7/2022	Section 4.1: Tag 64 FieldName updated to SettlementDate. Tag 151 Description updated with respect to Market Maker quotes.
1.1.1	07/23/2024	Updated SSL usage in our Certification environment.
1.1.2	08/09/2024	Notice added pertaining to duplicated trade message.